# 1c_bonus-E.md

```
Key update since “the old days”: AWS WAF logging can go direc
tly to CloudWatch Logs, S3, or Kinesis Data Firehose,
and you can associate one destination per Web ACL. Also, the
destination name must start with aws-waf-logs-.


Terraform supports this with aws_wafv2_web_acl_logging_config
uration.
Terraform Registry

Below is Lab 1C-Bonus-E (continued): WAF logging in Terraform
(with toggles), plus verification commands.

1) Add variables (append to variables.tf)
variable "waf_log_destination" {
  description = "Choose ONE destination per WebACL: cloudwatc
h | s3 | firehose"
  type        = string
  default     = "cloudwatch"
}

variable "waf_log_retention_days" {
  description = "Retention for WAF CloudWatch log group."
  type        = number
  default     = 14
}

variable "enable_waf_sampled_requests_only" {
  description = "If true, students can optionally filter/reda
ct fields later. (Placeholder toggle.)"
  type        = bool
  default     = false
```

```
}


2) Add file: bonus_b_waf_logging.tf (Look in Folder)

This provides three skeleton options (CloudWatch / S3 / Firehose). Students choose one via var.waf_log_destination.


3) Outputs (append to outputs.tf)
# Explanation: Coordinates for the WAF log destination—Chewbacca wants to know where the footprints landed.
output "chewbacca_waf_log_destination" {
  value = var.waf_log_destination
}

output "chewbacca_waf_cw_log_group_name" {
  value = var.waf_log_destination == "cloudwatch" ? aws_cloudwatch_log_group.chewbacca_waf_log_group01[0].name : null
}

output "chewbacca_waf_logs_s3_bucket" {
  value = var.waf_log_destination == "s3" ? aws_s3_bucket.chewbacca_waf_logs_bucket01[0].bucket : null
}

output "chewbacca_waf_firehose_name" {
  value = var.waf_log_destination == "firehose" ? aws_kinesis_firehose_delivery_stream.chewbacca_waf_firehose01[0].name : null
}


4) Student verification (CLI)
A) Confirm WAF logging is enabled (authoritative)
  aws wafv2 get-logging-configuration \
```

```
    --resource-arn <WEB_ACL_ARN>

Expected: LogDestinationConfigs contains exactly one destinat
ion.

B) Generate traffic (hits + blocks)
  curl -I https://chewbacca-growl.com/
  curl -I https://app.chewbacca-growl.com/

C1) If CloudWatch Logs destination
  aws logs describe-log-streams \
  --log-group-name aws-waf-logs-<project>-webacl01 \
  --order-by LastEventTime --descending

Then pull recent events:
  aws logs filter-log-events \
  --log-group-name aws-waf-logs-<project>-webacl01 \
  --max-items 20

C2) If S3 destination
  aws s3 ls s3://aws-waf-logs-<project>-<account_id>/ --recur
sive | head

C3) If Firehose destination
  aws firehose describe-delivery-stream \
  --delivery-stream-name aws-waf-logs-<project>-firehose01 \
  --query "DeliveryStreamDescription.DeliveryStreamStatus"

And confirm objects land:
  aws s3 ls s3://<firehose_dest_bucket>/waf-logs/ --recursive
| head

5) Why this makes incident response “real”
Now you can answer questions like:
  “Are 5xx caused by attackers or backend failure?”
  “Do we see WAF blocks spike before ALB 5xx?”
```

```
  “What paths / IPs are hammering the app?”
  “Is it one client, one ASN, one country, or broad?”
  “Did WAF mitigate, or are we failing downstream?”

This is precisely why WAF logging destinations include CloudWatch Logs (fast search) and S3/Firehose (archive/SIEM pipeline)
```

Based on my review of the Bonus E instructions and your current files, here's the breakdown:

---

# Lab 1C Bonus E: WAF Logging — Implementation Steps

## Duplicate Check Summary

| Item | File | Status |
|---|---|---|
| WAF logging variables | `03-variables.tf` | ✅ **None exist** — safe to add |
| WAF logging outputs | `05-outputs.tf` | ✅ **None exist** — safe to add |
| WAF logging resources | Main files | ✅ **None exist** — create new file |

**Note:** The instructions use `chewbacca_` naming, but your project uses `helga_`. Adapt accordingly.

---

## Step 1: Add Variables to `03-variables.tf`

Append these at the bottom:

```
# ============================================
# BONUS E: WAF Logging Variables
# ============================================
variable "waf_log_destination" {
  description = "Choose ONE destination per WebACL: cloudwatch | s3 | firehose"
  type        = string
  default     = "cloudwatch"
}
```

```
variable "waf_log_retention_days" {
  description = "Retention for WAF CloudWatch log group."
  type        = number
  default     = 14
}

variable "enable_waf_sampled_requests_only" {
  description = "If true, students can optionally filter/redact fields later. (Placeholder toggle.)"
  type        = bool
  default     = false
}
```

---

## Step 2: Create New File `04-1ce-Main.tf`

✅ **Prerequisite confirmed:** WAF Web ACL exists as `aws_wafv2_web_acl.helga_waf01\\\[0\\\]` in `04-1cb-Main.tf`.

Create this file with the following content (adapted from `bonus_b_waf_`logging.tf` reference):

```
############################################
# Bonus E - WAF Logging (CloudWatch Logs OR S3 OR Firehose)
# One destination per Web ACL, choose via var.waf_log_destination.
############################################

############################################
# Option 1: CloudWatch Logs destination
############################################

# Explanation: WAF logs in CloudWatch are your "blaster-cam footage"—fast search, fast triage, fast truth.
resource "aws_cloudwatch_log_group" "helga_waf_log_group01" {
  count = var.waf_log_destination == "cloudwatch" ? 1 : 0
```

```
  # NOTE: AWS requires WAF log destination names start with a
ws-waf-logs- (do not rename this).
  name              = "aws-waf-logs-${var.project_name}-webac
l01"
  retention_in_days = var.waf_log_retention_days

  tags = {
    Name = "${var.project_name}-waf-log-group01"
  }
}

# Explanation: This wire connects the shield generator to the
black box—WAF -> CloudWatch Logs.
resource "aws_wafv2_web_acl_logging_configuration" "helga_waf
_logging01" {
  count = var.enable_waf && var.waf_log_destination == "cloud
watch" ? 1 : 0

  resource_arn = aws_wafv2_web_acl.helga_waf01[0].arn
  log_destination_configs = [
    aws_cloudwatch_log_group.helga_waf_log_group01[0].arn
  ]

  # TODO: Add redacted_fields (authorization headers, cookie
s, etc.) as a stretch goal.
  # redacted_fields { ... }

  depends_on = [aws_wafv2_web_acl.helga_waf01]
}

############################################
# Option 2: S3 destination (direct)
############################################

# Explanation: S3 WAF logs are the long-term archive—receipts
```

```
that survive dashboards.
resource "aws_s3_bucket" "helga_waf_logs_bucket01" {
  count = var.waf_log_destination == "s3" ? 1 : 0

  bucket = "aws-waf-logs-${var.project_name}-${data.aws_calle
r_identity.current.account_id}"

  tags = {
    Name = "${var.project_name}-waf-logs-bucket01"
  }
}

# Explanation: Public access blocked—WAF logs are not a bedti
me story for the entire internet.
resource "aws_s3_bucket_public_access_block" "helga_waf_logs_
pab01" {
  count = var.waf_log_destination == "s3" ? 1 : 0

  bucket                  = aws_s3_bucket.helga_waf_logs_buck
et01[0].id
  block_public_acls       = true
  block_public_policy     = true
  ignore_public_acls      = true
  restrict_public_buckets = true
}

# Explanation: Connect shield generator to archive vault—WAF
-> S3.
resource "aws_wafv2_web_acl_logging_configuration" "helga_waf
_logging_s3_01" {
  count = var.enable_waf && var.waf_log_destination == "s3" ?
1 : 0

  resource_arn = aws_wafv2_web_acl.helga_waf01[0].arn
  log_destination_configs = [
    aws_s3_bucket.helga_waf_logs_bucket01[0].arn
```

```
  ]

  depends_on = [aws_wafv2_web_acl.helga_waf01]
}

############################################
# Option 3: Firehose destination (classic "stream then stor
e")
############################################

# Explanation: Firehose is the conveyor belt—WAF logs ride it
to storage (and can fork to SIEM later).
resource "aws_s3_bucket" "helga_firehose_waf_dest_bucket01" {
  count = var.waf_log_destination == "firehose" ? 1 : 0

  bucket = "${var.project_name}-waf-firehose-dest-${data.aws_
caller_identity.current.account_id}"

  tags = {
    Name = "${var.project_name}-waf-firehose-dest-bucket01"
  }
}

# Explanation: Firehose needs a role—don't let random droids
write into storage.
resource "aws_iam_role" "helga_firehose_role01" {
  count = var.waf_log_destination == "firehose" ? 1 : 0
  name  = "${var.project_name}-firehose-role01"

  assume_role_policy = jsonencode({
    Version = "2012-10-17"
    Statement = [{
      Effect = "Allow"
      Principal = { Service = "firehose.amazonaws.com" }
      Action = "sts:AssumeRole"
    }]
```

```
  })
}

# Explanation: Minimal permissions—allow Firehose to put obje
cts into the destination bucket.
resource "aws_iam_role_policy" "helga_firehose_policy01" {
  count = var.waf_log_destination == "firehose" ? 1 : 0
  name  = "${var.project_name}-firehose-policy01"
  role  = aws_iam_role.helga_firehose_role01[0].id

  policy = jsonencode({
    Version = "2012-10-17"
    Statement = [
      {
        Effect = "Allow"
        Action = [
          "s3:AbortMultipartUpload",
          "s3:GetBucketLocation",
          "s3:GetObject",
          "s3:ListBucket",
          "s3:ListBucketMultipartUploads",
          "s3:PutObject"
        ]
        Resource = [
          aws_s3_bucket.helga_firehose_waf_dest_bucket01[0].a
rn,
          "${aws_s3_bucket.helga_firehose_waf_dest_bucket01
[0].arn}/*"
        ]
      }
    ]
  })
}

# Explanation: The delivery stream is the belt itself—logs mo
ve from WAF -> Firehose -> S3.
```

```
resource "aws_kinesis_firehose_delivery_stream" "helga_waf_firehose01" {
  count       = var.waf_log_destination == "firehose" ? 1 : 0
  name        = "aws-waf-logs-${var.project_name}-firehose01"
  destination = "extended_s3"

  extended_s3_configuration {
    role_arn   = aws_iam_role.helga_firehose_role01[0].arn
    bucket_arn = aws_s3_bucket.helga_firehose_waf_dest_bucket01[0].arn
    prefix     = "waf-logs/"
  }
}

# Explanation: Connect shield generator to conveyor belt—WAF -> Firehose stream.
resource "aws_wafv2_web_acl_logging_configuration" "helga_waf_logging_firehose01" {
  count = var.enable_waf && var.waf_log_destination == "firehose" ? 1 : 0

  resource_arn = aws_wafv2_web_acl.helga_waf01[0].arn
  log_destination_configs = [
    aws_kinesis_firehose_delivery_stream.helga_waf_firehose01[0].arn
  ]

  depends_on = [aws_wafv2_web_acl.helga_waf01]
}
```

⚠️ **Note:** This file references `data.aws` `_caller_identity.current` which is defined in `04-1cd-Main.tf`. Ensure that data source is uncommented/active.

---

## Step 3: Add Outputs to `05-outputs.tf`

Append these at the bottom (adapted to `helga_` naming):

```
# ============================================
# BONUS E: WAF Logging Outputs
# ============================================
output "helga_waf_log_destination" {
  value = var.waf_log_destination
}

output "helga_waf_cw_log_group_name" {
  value = var.waf_log_destination == "cloudwatch" ? aws_cloudwatch_log_group.helga_waf_log_group01[0].name : null
}

output "helga_waf_logs_s3_bucket" {
  value = var.waf_log_destination == "s3" ? aws_s3_bucket.helga_waf_logs_bucket01[0].bucket : null
}

output "helga_waf_firehose_name" {
  value = var.waf_log_destination == "firehose" ? aws_kinesis_firehose_delivery_stream.helga_waf_firehose01[0].name : null
}
```

---

## Step 4: Verification Commands

After `terraform apply`:

**A) Confirm WAF logging is enabled:**

```
aws wafv2 get-logging-configuration \
  --resource-arn <WEB_ACL_ARN>
```

```
aws wafv2 get-logging-configuration \
  --resource-arn arn:aws:wafv2:us-east-2:919113286081:regional/webacl/helga-waf01/719d7e8f-9d85-476e-b5a4-69aeaa95922f
```

**B) Generate traffic:**

```
aws logs filter-log-events \
--log-group-name aws-waf-logs-helga-webacl01 \
--max-items 5
```

**C) Check logs based on destination:**

- **CloudWatch:** `aws logs filter-log-events --log-group-name aws-waf-logs-helga-webacl01 --max-items 20`
- **S3:** `aws s3 ls s3://aws-waf-logs-helga-\\\\\\<account_id\\\\\\>/ --recursive \\\\\\| head`
- **Firehose:** `aws firehose describe-delivery-stream --delivery-stream-name aws-waf-logs-helga-firehose01`

---

**Next step:** Do you want me to check `04-1cb-Main.tf` to confirm the WAF Web ACL exists, or would you like me to draft the full `04-1ce-Main.tf` file with the WAF logging resources?

**Yes — Bonus E is complete.** ✅

Your logs confirm everything is working:

| Checkpoint | Status |
|---|---|
| WAF logging enabled | ✅ `LogDestinationConfigs` points to CloudWatch |
| Logs flowing | ✅ 5 events returned |
| ALB attached | ✅ `httpSourceName: ALB`, `helga-alb01` |
| Managed rules evaluated | ✅ `AWSManagedRulesCommonRuleSet` in `ruleGroupList` |

---

## Bonus observation: You're already being scanned

Your logs show requests from:

- **France** (`91.231.89.120`) — hit `williebright.com`
- **Portugal** (`185.180.140.112`) — hit the raw IP `18.216.38.254`

- **US** (`170.106.163.84`) — also hit the raw IP, with `zh-CN` language headers (likely a cloud-hosted scanner)

All were **ALLOWED** (action: `ALLOW`, terminatingRuleId: `Default_Action`) because they didn't trigger any WAF rules. This is normal internet background noise — and exactly what WAF logging is designed to surface.

---

## What's next?

If you want to see a **BLOCK** in the logs (to prove your WAF actually stops bad traffic), you can try triggering the `AWSManagedRulesCommonRuleSet` with a known-bad request pattern, like:

```
curl -I "
```

Then re-run the log query — you should see `action: BLOCK`.

Otherwise, you're done with Bonus E. Ready for the next section?


```
        {
            "logStreamName": "us-east-2_helga-waf01_0",
            "timestamp": 1768625523260,
            "message": "{\"timestamp\":1768625523260,\"formatVersion\":1,\"webaclId\":\"arn:aws:wafv2:us-ea
t-2:919113286081:regional/webacl/helga-waf01/719d7e8f-9d85-476e-b5a4-69aeaa95922f\",\"terminatingRuleId\":\
Default_Action\",\"terminatingRuleType\":\"REGULAR\",\"action\":\"ALLOW\",\"terminatingRuleMatchDetails\":[
,\"httpSourceName\":\"ALB\",\"httpSourceId\":\"919113286081-app/helga-alb01/4cc4f3a53980daa6\",\"ruleGroupL
st\":[{\"ruleGroupId\":\"AWS#AWSManagedRulesCommonRuleSet\",\"terminatingRule\":null,\"nonTerminatingMatchi
gRules\":[],\"excludedRules\":null,\"customerConfig\":null}],\"rateBasedRuleList\":[],\"nonTerminatingMatch
ngRules\":[],\"requestHeadersInserted\":null,\"responseCodeSent\":null,\"httpRequest\":{\"clientIp\":\"185.
80.140.112\",\"country\":\"PT\",\"headers\":[{\"name\":\"Host\",\"value\":\"18.216.38.254\"},{\"name\":\"Us
r-Agent\",\"value\":\"Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chro
e/60.0.3112.113 Safari/537.36\"},{\"name\":\"Accept\",\"value\":\"*/*\"},{\"name\":\"Referer\",\"value\":\"
ttp://18.216.38.254/\"},{\"name\":\"Accept-Encoding\",\"value\":\"gzip\"}],\"uri\":\"/\",\"args\":\"\",\"ht
pVersion\":\"HTTP/1.1\",\"httpMethod\":\"GET\",\"requestId\":\"1-696b1573-06d984a76982b8467bc354e9\",\"frag
ent\":\"\",\"scheme\":\"https\",\"host\":\"18.216.38.254\"},\"ja3Fingerprint\":\"cba7f34191ef2379c1325641f6
6c4f4\",\"ja4Fingerprint\":\"t12i130500_2d7513195f68_021165082e1c\"}",
            "ingestionTime": 1768625542832,
            "eventId": "39441667145839155938970611853902303537645484737346273280"
```

```
  --log-group-name aws-waf-logs-helga-webacl01 \
  --max-items 5
```


```
on/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=
b3;q=0.7\"},{\"name\":\"Accept-Encoding\",\"value\":\"gzip\"},{\"name\":\"Accept-Language\",\"value\":\"zh-C
N,zh;q=0.9,en-US;q=0.8,en;q=0.7\"},{\"name\":\"Cache-Control\",\"value\":\"no-cache\"},{\"name\":\"Connectio
n\",\"value\":\"keep-alive\"},{\"name\":\"Pragma\",\"value\":\"no-cache\"},{\"name\":\"Referer\",\"value\":\
"http://18.216.38.254\"},{\"name\":\"Upgrade-Insecure-Requests\",\"value\":\"1\"},{\"name\":\"Connection\",\
"value\":\"close\"}],\"uri\":\"/\",\"args\":\"\",\"httpVersion\":\"HTTP/1.1\",\"httpMethod\":\"GET\",\"reque
stId\":\"1-696b1601-441e5ed031d432bb71911ae7\",\"fragment\":\"\",\"scheme\":\"https\",\"host\":\"18.216.38.2
54\"},\"ja3Fingerprint\":\"19e29534fd49dd27d09234e639c4057e\",\"ja4Fingerprint\":\"t13i190800_9dc949149365_9
7f8aa674fd9\"}",
            "ingestionTime": 1768625684218,
            "eventId": "39441670313615409899848568033693394550815714044573253632"
        }
    ],
    "searchedLogStreams": [],
    "NextToken": "eyJuZXh0VG9rZW4iOiAiQnhrcTZrVkdGdHEyeV9Nb2lnZXFzY1BPZGhYVmJoaVZ0TG9BbVhiNWpDb2NvdTZnQWUxeH
M3VWdFaXJDZXNHNV9fUWFDSktQanY0NUVmUGtNS0o3X3JyamtSdTJQTDBhRG9SY25jcEIxTjdyVWxnQzdQdVk4R2RlUjRqdFl2UzhsV1VDUV
dkOEttbXJEVk5xT2dxQTNoTkxmZ21ZdnJLUjNjX0F6X2F0T2FKWUVOaWFBVTZITUZlcFZ3a01rVkkzOG50MnVIYUF5Y2VWVWhSaFBuWmJ4Rk
JFamVoUDhTNzY4UkUtY0dvMjdYSW8wb2pxOTh2RFpPQk0zbkJOUC1HVkxPQTNVMUtzYUU3TzZSNmVEamlBczA3amp2TExkTlNaWFFMeFlVN0
xDdW9HSWswcUJ1cElrNDB1dFhyWU5VekNQU1ZYaWJmZlFObm51b2dhYmVUOXl3In0="
}
```



```
            "timestamp": 1768625665308,
            "message": "{\"timestamp\":1768625665308,\"formatVersion\":1,\"webaclId\":\"arn:aws:wafv2:us-eas
t-2:919113286081:regional/webacl/helga-waf01/719d7e8f-9d85-476e-b5a4-69aeaa95922f\",\"terminatingRuleId\":\"
Default_Action\",\"terminatingRuleType\":\"REGULAR\",\"action\":\"ALLOW\",\"terminatingRuleMatchDetails\":[]
,\"httpSourceName\":\"ALB\",\"httpSourceId\":\"919113286081-app/helga-alb01/4cc4f3a53980daa6\",\"ruleGroupLi
st\":[{\"ruleGroupId\":\"AWS#AWSManagedRulesCommonRuleSet\",\"terminatingRule\":null,\"nonTerminatingMatchin
gRules\":[],\"excludedRules\":null,\"customerConfig\":null}],\"rateBasedRuleList\":[],\"nonTerminatingMatchi
ngRules\":[],\"requestHeadersInserted\":null,\"responseCodeSent\":null,\"httpRequest\":{\"clientIp\":\"170.1
06.163.84\",\"country\":\"US\",\"headers\":[{\"name\":\"Host\",\"value\":\"18.216.38.254\"},{\"name\":\"User
-Agent\",\"value\":\"Mozilla/5.0 (iPhone; CPU iPhone OS 13_2_3 like Mac OS X) AppleWebKit/605.1.15 (KHTML, l
ike Gecko) Version/13.0.3 Mobile/15E148 Safari/604.1\"},{\"name\":\"Accept\",\"value\":\"text/html,applicati
on/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=
b3;q=0.7\"},{\"name\":\"Accept-Encoding\",\"value\":\"gzip\"},{\"name\":\"Accept-Language\",\"value\":\"zh-C
N,zh;q=0.9,en-US;q=0.8,en;q=0.7\"},{\"name\":\"Cache-Control\",\"value\":\"no-cache\"},{\"name\":\"Connectio
n\",\"value\":\"keep-alive\"},{\"name\":\"Pragma\",\"value\":\"no-cache\"},{\"name\":\"Referer\",\"value\":\
"http://18.216.38.254\"},{\"name\":\"Upgrade-Insecure-Requests\",\"value\":\"1\"},{\"name\":\"Connection\",\
"value\":\"close\"}],\"uri\":\"/\",\"args\":\"\",\"httpVersion\":\"HTTP/1.1\",\"httpMethod\":\"GET\",\"reque
stId\":\"1-696b1601-441e5ed031d432bb71911ae7\",\"fragment\":\"\",\"scheme\":\"https\",\"host\":\"18.216.38.2
54\"},\"ja3Fingerprint\":\"19e29534fd49dd27d09234e639c4057e\",\"ja4Fingerprint\":\"t13i190800_9dc949149365_9
7f8aa674fd9\"}",
            "ingestionTime": 1768625684218,
```

```
            "logStreamName": "us-east-2_helga-waf01_0",
            "timestamp": 1768625664270,
            "message": "{\"timestamp\":1768625664270,\"formatVersion\":1,\"webaclId\":\"arn:aws:wafv2:us-eas
t-2:919113286081:regional/webacl/helga-waf01/719d7e8f-9d85-476e-b5a4-69aeaa95922f\",\"terminatingRuleId\":\"
Default_Action\",\"terminatingRuleType\":\"REGULAR\",\"action\":\"ALLOW\",\"terminatingRuleMatchDetails\":[]
,\"httpSourceName\":\"ALB\",\"httpSourceId\":\"919113286081-app/helga-alb01/4cc4f3a53980daa6\",\"ruleGroupLi
st\":[{\"ruleGroupId\":\"AWS#AWSManagedRulesCommonRuleSet\",\"terminatingRule\":null,\"nonTerminatingMatchin
gRules\":[],\"excludedRules\":null,\"customerConfig\":null}],\"rateBasedRuleList\":[],\"nonTerminatingMatchi
ngRules\":[],\"requestHeadersInserted\":null,\"responseCodeSent\":null,\"httpRequest\":{\"clientIp\":\"170.1
06.163.84\",\"country\":\"US\",\"headers\":[{\"name\":\"Host\",\"value\":\"18.216.38.254\"},{\"name\":\"User
-Agent\",\"value\":\"Mozilla/5.0 (iPhone; CPU iPhone OS 13_2_3 like Mac OS X) AppleWebKit/605.1.15 (KHTML, l
ike Gecko) Version/13.0.3 Mobile/15E148 Safari/604.1\"},{\"name\":\"Accept\",\"value\":\"text/html,applicati
on/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=
b3;q=0.7\"},{\"name\":\"Accept-Encoding\",\"value\":\"gzip\"},{\"name\":\"Accept-Language\",\"value\":\"zh-C
N,zh;q=0.9,en-US;q=0.8,en;q=0.7\"},{\"name\":\"Cache-Control\",\"value\":\"no-cache\"},{\"name\":\"Connectio
n\",\"value\":\"keep-alive\"},{\"name\":\"Pragma\",\"value\":\"no-cache\"},{\"name\":\"Upgrade-Insecure-Requ
ests\",\"value\":\"1\"},{\"name\":\"Connection\",\"value\":\"close\"}],\"uri\":\"/\",\"args\":\"\",\"httpVer
sion\":\"HTTP/1.1\",\"httpMethod\":\"GET\",\"requestId\":\"1-696b1600-141a892020be94df60efb188\",\"fragment\
":\"\",\"scheme\":\"http\",\"host\":\"18.216.38.254\"}}",
            "ingestionTime": 1768625675376,
            "eventId": "39441670290467236383773781202090173089575433001399091200"
```

```
        {
            "logStreamName": "us-east-2_helga-waf01_0",
            "timestamp": 1768625523025,
            "message": "{\"timestamp\":1768625523025,\"formatVersion\":1,\"webaclId\":\"arn:aws:wafv2:us-eas
t-2:919113286081:regional/webacl/helga-waf01/719d7e8f-9d85-476e-b5a4-69aeaa95922f\",\"terminatingRuleId\":\"
Default_Action\",\"terminatingRuleType\":\"REGULAR\",\"action\":\"ALLOW\",\"terminatingRuleMatchDetails\":[]
,\"httpSourceName\":\"ALB\",\"httpSourceId\":\"919113286081-app/helga-alb01/4cc4f3a53980daa6\",\"ruleGroupLi
st\":[{\"ruleGroupId\":\"AWS#AWSManagedRulesCommonRuleSet\",\"terminatingRule\":null,\"nonTerminatingMatchin
gRules\":[],\"excludedRules\":null,\"customerConfig\":null}],\"rateBasedRuleList\":[],\"nonTerminatingMatchi
ngRules\":[],\"requestHeadersInserted\":null,\"responseCodeSent\":null,\"httpRequest\":{\"clientIp\":\"185.1
80.140.112\",\"country\":\"PT\",\"headers\":[{\"name\":\"Host\",\"value\":\"18.216.38.254\"},{\"name\":\"Use
r-Agent\",\"value\":\"Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrom
e/60.0.3112.113 Safari/537.36\"},{\"name\":\"Accept\",\"value\":\"*/*\"},{\"name\":\"Accept-Encoding\",\"val
ue\":\"gzip\"}],\"uri\":\"/\",\"args\":\"\",\"httpVersion\":\"HTTP/1.1\",\"httpMethod\":\"GET\",\"requestId\
":\"1-696b1573-2b9fea574fd95c2b068ba5d4\",\"fragment\":\"\",\"scheme\":\"http\",\"host\":\"18.216.38.254\"}}
",
            "ingestionTime": 1768625533937,
            "eventId": "39441667140598480817315915404888074218187472958908989440"
        },
```

```
$ aws logs filter-log-events \
  --log-group-name aws-waf-logs-helga-webacl01 \
  --max-items 5
{
    "events": [
        {
            "logStreamName": "us-east-2_helga-waf01_0",
            "timestamp": 1768625455257,
            "message": "{\"timestamp\":1768625455257,\"formatVersion\":1,\"webaclId\":\"arn:aws:wafv2:us-eas
t-2:919113286081:regional/webacl/helga-waf01/719d7e8f-9d85-476e-b5a4-69aeaa95922f\",\"terminatingRuleId\":\"
Default_Action\",\"terminatingRuleType\":\"REGULAR\",\"action\":\"ALLOW\",\"terminatingRuleMatchDetails\":[]
,\"httpSourceName\":\"ALB\",\"httpSourceId\":\"919113286081-app/helga-alb01/4cc4f3a53980daa6\",\"ruleGroupLi
st\":[{\"ruleGroupId\":\"AWS#AWSManagedRulesCommonRuleSet\",\"terminatingRule\":null,\"nonTerminatingMatchin
gRules\":[],\"excludedRules\":null,\"customerConfig\":null}],\"rateBasedRuleList\":[],\"nonTerminatingMatchi
ngRules\":[],\"requestHeadersInserted\":null,\"responseCodeSent\":null,\"httpRequest\":{\"clientIp\":\"91.23
1.89.120\",\"country\":\"FR\",\"headers\":[{\"name\":\"Host\",\"value\":\"williebright.com\"},{\"name\":\"Co
nnection\",\"value\":\"close\"},{\"name\":\"User-Agent\",\"value\":\"Mozilla/5.0 (X11; Ubuntu; Linux x86_64;
 rv:134.0) Gecko/20100101 Firefox/134.0\"},{\"name\":\"Accept\",\"value\":\"text/html,application/xhtml+xml,
application/xml;q=0.9,*/*;q=0.8\"},{\"name\":\"Accept-Language\",\"value\":\"en-US,en;q=0.5\"}],\"uri\":\"/\
",\"args\":\"\",\"httpVersion\":\"HTTP/1.1\",\"httpMethod\":\"GET\",\"requestId\":\"1-696b152f-1efbf046528a7
59d09588e49\",\"fragment\":\"\",\"scheme\":\"http\",\"host\":\"williebright.com\"}}",
            "ingestionTime": 1768625476477,
            "eventId": "39441665629321580203292646279830549084682392518976274432"
```

```
brigh@DESKTOP-62DEPP9 MINGW64 ~/Documents/TheoWAF/class7/AWS/Classes/armageddon_3 (master)
$ aws wafv2 get-logging-configuration \
  --resource-arn arn:aws:wafv2:us-east-2:919113286081:regional/webacl/helga-waf01/719d7e8f-9d85-476e-b5a4-69
aeaa95922f
{
    "LoggingConfiguration": {
        "ResourceArn": "arn:aws:wafv2:us-east-2:919113286081:regional/webacl/helga-waf01/719d7e8f-9d85-476e-
b5a4-69aeaa95922f",
        "LogDestinationConfigs": [
            "arn:aws:logs:us-east-2:919113286081:log-group:aws-waf-logs-helga-webacl01"
        ],
        "ManagedByFirewallManager": false,
        "LogType": "WAF_LOGS",
        "LogScope": "CUSTOMER"
    }
}
```

---

# ✅ Bonus E Complete — Summary

**Completed:** January 17, 2026

## Objectives Achieved

| Objective | Status |
| --- | --- |
| WAF logging configuration enabled | ✅ Complete |
| CloudWatch Logs destination configured | ✅ Complete |
| Logs flowing from WAF → CloudWatch | ✅ Verified |

| Objective | Status |
| --- | --- |
| Managed rules ( `AWSManagedRulesCommonRuleSet` ) evaluating | ✅ Confirmed |

## Files Created/Modified

- `03-variables.tf` — Added `waf_log_destination`, `waf_log_retention_days`, `enable_waf_sampled_requests_only`
- `04-lce-Main.tf` — Created WAF logging resources (CloudWatch, S3, Firehose options)
- `05-outputs.tf` — Added WAF logging outputs

## Key Learnings

1. `ManagedByFirewallManager: false` — This is expected when you create WAF directly with Terraform (not through AWS Firewall Manager)
2. **Destination naming constraint** — AWS requires WAF log destinations to start with `aws-waf-logs-`
3. **One destination per Web ACL** — You can choose CloudWatch, S3, or Firehose, but only one at a time

## Final Verification Output

```
aws wafv2 get-logging-configuration \
  --resource-arn arn:aws:wafv2:us-east-2:919113286081:regiona
l/webacl/helga-waf01/719d7e8f-9d85-476e-b5a4-69aeaa95922f

# Result:
{
  "LoggingConfiguration": {
    "ResourceArn": "arn:aws:wafv2:us-east-2:919113286081:regi
onal/webacl/helga-waf01/...",
    "LogDestinationConfigs": [
      "arn:aws:logs:us-east-2:919113286081:log-group:aws-waf-
logs-helga-webacl01"
    ],
```

```
    "LogType": "WAF_LOGS",
    "LogScope": "CUSTOMER"
  }
}
```

```
aws logs filter-log-events \
  --log-group-name aws-waf-logs-helga-webacl01 \
  --max-items 5

# Result: 5 events returned with action: ALLOW, httpSourceNam
e: ALB
```

### Bonus Observation: Real-World Traffic Captured

WAF logs immediately surfaced background internet noise:

- **France** (`91.231.89.120`) — hit `williebright.com`
- **Portugal** (`185.180.140.112`) — hit raw IP `18.216.38.254`
- **US** (`170.106.163.84`) — cloud-hosted scanner with `zh-CN` headers

All requests were **ALLOWED** (no rule violations). This is exactly what WAF logging is designed to surface for incident response.

**Result:** WAF logging operational. Full request visibility achieved for security monitoring and incident response.